

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-0D7E8A5E 0d7e8a5e-36e4-4e8c-8146-d92b56b2b440
Submitted by	testreviewer L!
Submission date	2026-10-07 06:44 UTC
Documents	1 Types: Payment evidence
Case type	School document
Status / decision	Pending manual review / Pending review
Report generated	2026-10-07 06:52 UTC by testreviewer L! (Reviewer L1)
Report ID	e0d9ea29-a704-4721-84fd-12802f01c71d

2. Executive Summary

LOW RISK Score 25 / 100 (computed 2026-10-07 06:50 UTC)

Why: Driven mainly by (25 points): 'case13School educational document.pdf': 1 localized area(s) show higher recompression error than the rest of the page (error level analysis) — a possible sign of editing. First on page 1.

Decision status: Pending review.

Exceptions: 3 item(s) did not pass, 3 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The risk engine is a transparent, weighted rule set. Each row is one rule that fired, with the reason text stored at scoring time. Points are summed (raw 25), capped at 100, and mapped to a tier (medium from 30, high from 60) — here **25/100**, **low**.

Severity	Points	Check	Document / page	Reason
MEDIUM	25	Error level analysis (ELA)	case13School educational document.pdf · page 1	'case13School educational document.pdf': 1 localized area(s) show higher recompression error than the rest of the page (error level analysis) — a possible sign of editing. First on page 1. Highlights: R3

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — case13School educational document.pdf (Payment evidence · 6 passed, 2 flagged)

Check	Result	Details
Metadata forensics PDF metadata, edit history and incremental-save markers.	Pass	No anomaly detected by this check.
Font consistency Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).	Flag	2 finding(s). Page 1 (scanned): 'require' looks like a different font (serif) from the text around it (humanist sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like...
Error level analysis (ELA) Pixel-level recompression differences that can indicate edited regions.	Flag	1 finding(s). Page 1: a localized area shows a higher JPEG recompression error than the rest of the page — a possible sign this area was edited after the page was last saved. This is a probabilistic signal, not a definitive finding; ordinary scan/print artifa...
Copy-move detection Regions of a page that were duplicated elsewhere on the same page.	Pass	No anomaly detected by this check.
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Pass	No anomaly detected by this check.
Signature / stamp detection Whether a signature or stamp mark is present where this kind of document normally carries one.	Pass	2 signature/stamp region(s) located (presence only).
Duplicate detection Perceptual-hash comparison of each page against previously submitted pages.	Pass	No anomaly detected by this check.
Field validation Date, subtotal / tax / total and reference-number plausibility.	Pass	7 sub-check(s) passed; none flagged.
Issuer verification Issuer name matched against the issuer registry.	Not checked	Not checked: the issuer registry has no active entries for in AE, so there was nothing to compare this issuer against.

Linked cases (reviewer note, not scored)

Other cases of this company that share a scanning device, a parent/guardian or a family/student ID with this one. Shared office scanners are common, so this is context for the reviewer only; it does not affect the risk score.

Case	Shared
CASE-866F937F	same scanning device (65C-1i, serial 0050AA53A8E2); same parent/guardian ('Shankar Ram Thodla (father's name in Sai Shankar Ram Thodla)' / 'Shankar Ram Thodla (father's name in Sai Shankar Ram Thodla)'); same student ID (083350742180)

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (3 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where	Highlight
1	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): 'require' looks like a different font (serif) from the text around it (humanist sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition...	Document 1 (case13School educational document.pdf) · page 1	R1
2	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): 'a' looks like a different font (serif) from the text around it (humanist sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone...	Document 1 (case13School educational document.pdf) · page 1	R2
3	ELA: possible tamper region Page 1: a localized area shows a higher JPEG recompression error than the rest of the page — a possible sign this area was edited after the page was last saved. This is a probabilistic signal, not a definitive finding; ordinary scan/print artifacts can also trigger it.	Document 1 (case13School educational document.pdf) · page 1	R3

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Ran	Ran on 1 of 1 document(s); 0 flagged.
Font consistency	Ran	Ran on 1 of 1 document(s); 1 flagged.
Error level analysis (ELA)	Ran	Ran on 1 of 1 document(s); 1 flagged.
Copy-move detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 0 flagged.
Signature / stamp detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Duplicate detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Field validation	Ran	Ran on 1 of 1 document(s); 0 flagged.
Issuer verification	Ran	Ran on 1 of 1 document(s); 0 flagged.
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

The risk score in this report (**25/100, low**) was computed at 2026-10-07 06:50 UTC (assessment 92247995-d902-49d1-8808-fccc92ca5caf) from the risk-rule versions that fired: ela.tamper_region_detected v1 (weight 25). Tier thresholds at that time: medium from 30, high from 60.

Rule weights are versioned and are never re-applied retroactively: if an administrator later edits a rule, this case is not re-scored, and **this report remains an accurate historical record** of what was known and weighted when it was generated. A report generated later may legitimately differ.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- The issuer registry (0 active entries) was assumed to be current and accurate as of 2026-10-07 06:52 UTC. An issuer that is absent from it is reported as “not in registry”, which is a prompt to verify the issuer — not a finding that the issuer is fraudulent.
- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: case13School educational document.pdf: Gateway transaction id (90% confidence); case13School educational document.pdf: Gateway transaction id continued (90% confidence).
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.



7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-07 06:44 UTC	Case created	testreviewer L! (Reviewer L1)	
2026-10-07 06:44 UTC	Document uploaded	testreviewer L! (Reviewer L1)	case13School educational document.pdf; SHA-256 bd91d951965ed892...
2026-10-07 06:47 UTC	Font consistency check completed	System	case13School educational document.pdf
2026-10-07 06:47 UTC	Text extraction, classification and field extraction completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	Signature / stamp detection completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	Visual inconsistency review completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	Field validation and issuer verification completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	Metadata forensics completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	Duplicate detection completed	System	case13School educational document.pdf
2026-10-07 06:49 UTC	ELA and copy-move checks completed	System	case13School educational document.pdf
2026-10-07 06:50 UTC	Risk score computed	System	tier low, score 25
2026-10-07 06:50 UTC	Case status changed	System	submitted → pending_manual_review; automated checks complete

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — case13School educational document.pdf

Field	Value	Confidence	Located
Issuer	The British International School Abu Dhabi	95%	p.1
Reference number	BAB010689	95%	p.1
Date	2026-08-03	95%	p.1
Amount	78,407.00 AED	95%	p.1
Subtotal	Not found	uncertain	
Tax rate	Not found	uncertain	
Tax amount	Not found	uncertain	
Address	The British International School Abu Dhabi, PO Box 60968, Abu Dhabi, United Arab Emirates	95%	p.1
Trn	100364873800003	95%	p.1
Phone	+971 2 510 0140	95%	p.1
Student name	Sai Shankar Ram Thodla	95%	p.1
Student id	083350742180	95%	p.1
Zenda transaction id	P150710B559	95%	p.1
Gateway transaction id	93cef9e6-1241-492c-978c-	90% uncertain	p.1
Gateway transaction id continued	5ee532581e58	90% uncertain	p.1
Payment made by	Sai Shankar Ram Thodla	95%	p.1
Payment note	This is a system generated document and does not require a signature and all payments are made to the school's account.	95%	
Document print time	3 August 2026 11:25:00 AM	95%	p.1

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256	Verification
case13School educational document.pdf 35,762 bytes	bd91d951965ed892e65d462470993f3dc6984b915e63a1e793742e1900936815	VERIFIED

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
Colour model	RGB, converted to BGR for OpenCV

Metadata forensics

Libraries	pikepdf / PyMuPDF (PDF structure, XMP, incremental updates)
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Modified-after-created rule	flagged if ModDate is more than 1 second(s) after CreationDate
Editing-software names matched	photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape

Error level analysis (ELA)

JPEG recompression quality	75
Error gain (scale)	50 (sqrt error term; the reference tool's default)
Contrast stretch / LUT	not applied — the raw error magnitude is thresholded directly
Region threshold	page-relative percentile 99.7 with an absolute floor of 60/255
Region area accepted	0.2% – 60% of the page
Applies to	pages with an embedded raster image only (born-digital text pages are exempt)

Copy-move detection

Keypoint algorithm	BRISK (OpenCV), detector threshold 200
Response / matching / distance thresholds	90 / 20 / 15
Minimum cluster size	30 matching keypoint pairs
Minimum region area	2% of the page
Keypoint / match caps	4,000 / 4,000

Duplicate detection

Hash	perceptual hash (imagehash pHash), 64-bit, one per rendered page
Hamming-distance threshold	5 (≤ this is flagged)
Compared against	every page hash of every previously submitted document

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge ≤ 1536px

Signature / stamp detection and comparison

Model	gpt-4.1-mini
Pages analysed per document	first 10
Output	presence and placement only; comparison is a qualitative verdict, never a numeric score

Field validation

Sub-checks	date not in the future; subtotal vs line items; total vs subtotal + tax; reference-number format
Amount tolerance	1% of the amount, minimum 0.01

Issuer verification

Matching	rapidfuzz weighted ratio against the active issuer registry, with a model-judged fallback for cross-script names
Similarity threshold	85

Field locations (purple highlights)

Method	extracted values matched back onto OCR word positions (Azure Document Intelligence Layout)
Repeated values	resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)

Accuracy	approximate; a field that cannot be matched is reported without a highlight
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9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

	ELA (solid red)
	Copy-move (solid orange)
	Font mismatch in the text layer (solid fuchsia)
	Field exception (solid purple)
	Model judgment (dashed blue, approximate)
	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (case13School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R2	Document 1 (case13School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R3	Document 1 (case13School educational document.pdf)	1	ELA — possible tamper region

case13School educational document.pdf — page 1

Section 9 · Document 1 of 1 · regions R1, R2, R3

ELA (solid red)

Font mismatch in the text layer (solid fuchsia)



**THE BRITISH
INTERNATIONAL SCHOOL
ABU DHABI**

RECEIPT

The British International School Abu Dhabi,
PO Box 60968, Abu Dhabi,
United Arab Emirates
TRN: 100364873800003
Phone: +971 2 510 0140

Sai Shankar Ram Thodla

Receipt Number: BAB010689
Date: 03-08-2026
Student ID: 083350742180
zenda transaction ID: P150710B559
Gateway transaction ID: 93cef9e6-1241-492c-978c-5ee532581e58

Particulars	Amount
Student Thodla (Student ID : 083350742180)	
26/27 - TERM 1 TUITION Y04 FEE, Aug/2026 (FY27-T1-47345)	44,804.00
26/27 - TERM 2 TUITION Y04 FEE, Dec/2026 (FY27-T2-49562)	33,603.00
Total Received	AED 78,407.00



R3 · ELA — possible tamper region



R2 · Font mismatch — text set in a different font family (scan, estimated)
R1 · Font mismatch — text set in a different font family (scan, estimated)

This payment was made on zenda by Sai Shankar Ram Thodla. This is a system generated document and does not require a signature and all payments are made to the school's account.

3 August 2026 11:25:00 AM

R1. Font consistency

Page 1 (scanned): 'require' looks like a different font (serif) from the text around it (humanist sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone is not reliable on a word or two.

R2. Font consistency

Page 1 (scanned): 'a' looks like a different font (serif) from the text around it (humanist sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone is not reliable on a word or two.

R3. Error level analysis (ELA)

Page 1: a localized area shows a higher JPEG recompression error than the rest of the page — a possible sign this area was edited after the page was last saved. This is a probabilistic signal, not a definitive finding; ordinary scan/print artifacts can also trigger it.